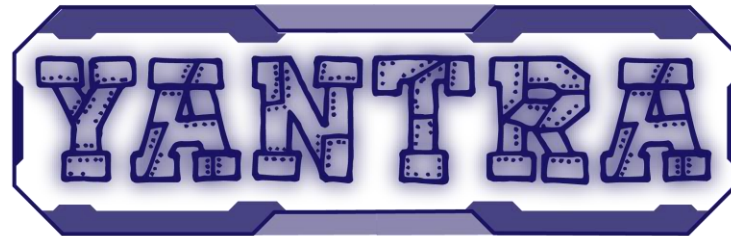


RAPID MANUFACTURING PROCESS OPTIMIZATION

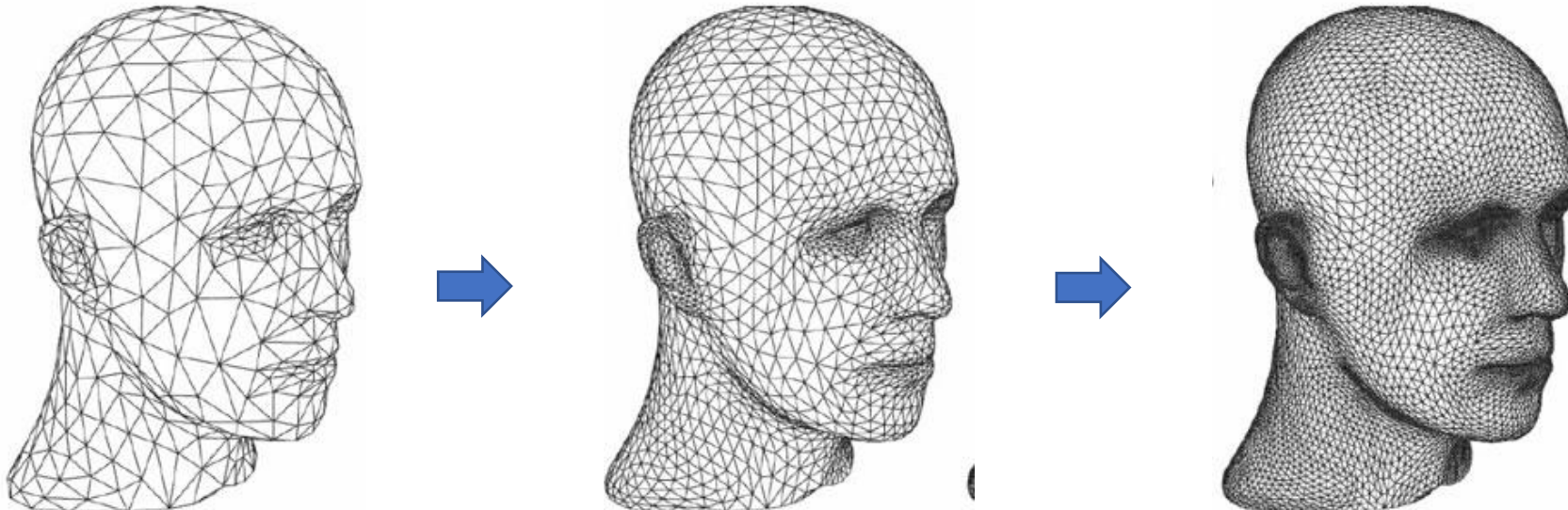


Factors Influencing Accuracy

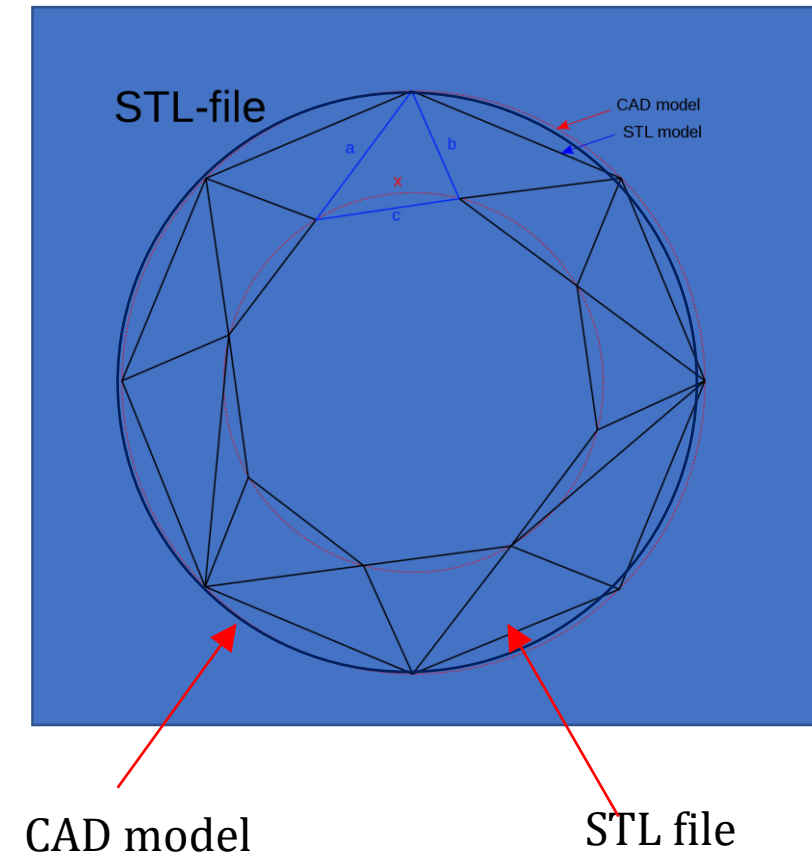
- As accuracy is Function of many different factors
- Some factors are interdependent.
- Thus, the **factors** that most influence RP process accuracy is classified into 3 categories.
 - **Factors causing errors during the data preparation** such as STL file generation, model slicing and part build direction.
 - **Factors influencing the part accuracy during the build stage** such as process specific parameters.
 - **Factors which are directly related to the part finishing techniques** such as postprocessing steps.

DATA PREPARATION ERRORS (ERROR DUE TO TESSELLATION)

- Most of the RP systems employ standard STL input files.
- A STL file approximates the surface of the 3D CAD model by triangles.
- Errors caused by tessellation are usually ignored because of the belief that tessellation errors can be minimized by **increasing the number of triangles**.



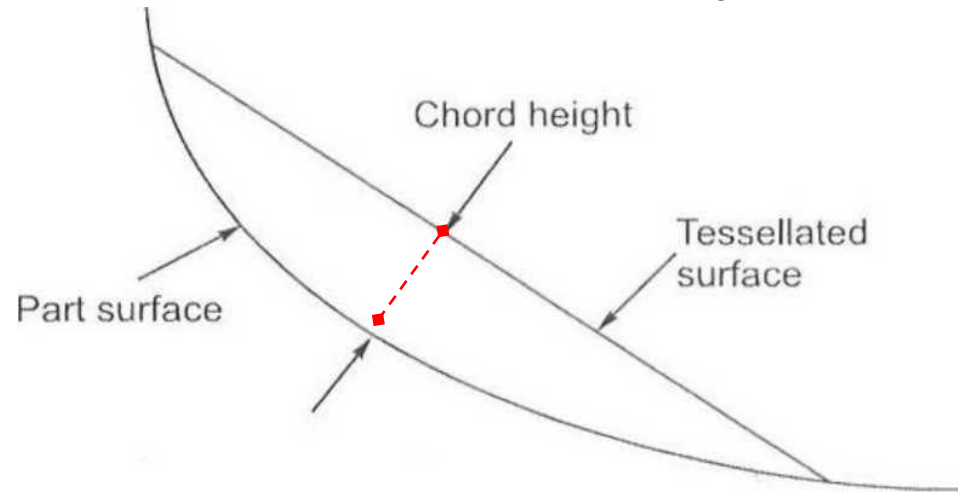
- However, in practice the number of triangles **cannot be increased indefinitely**. The resolution of STL files can be controlled during their generation in a 3D CAD system through tessellation parameters.
- The STL generation process can be controlled by specifying
 - Chord height
 - Angle control factor



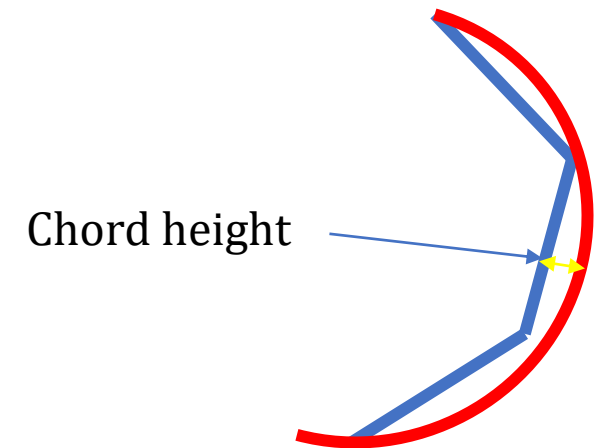
Chord height for surface tessellation

Chord Height:

- Specifies the maximum distance between a chord and surface.
- If **less deviation** from the actual part surface is required, a **smaller chord height** should be specified.
- The upper bound depends on the model size and lower bound for these parameters is a function of the CAD model accuracy.



Chord height for surface tessellation

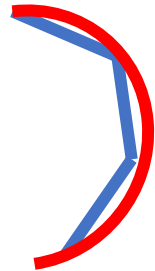


Angle Control for surface tessellation

Angle Control:

Specifies the required definition level along curves with small radius.

"It defines a **threshold** for the **curve radius** (r_0) below which the curves should be **tessellated**".

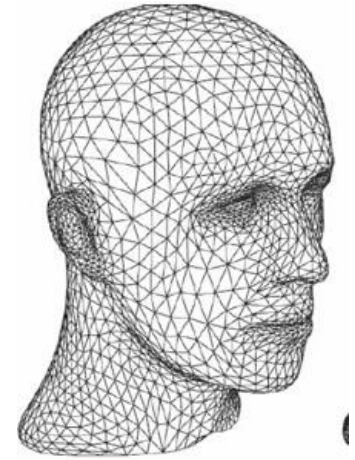


Smaller radius
should be tessellated

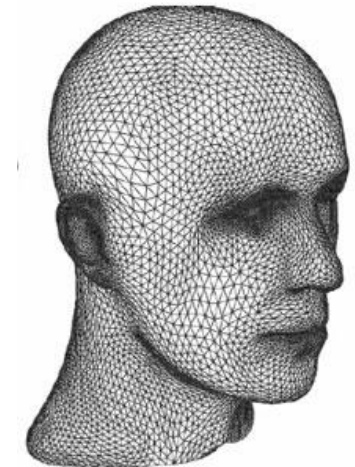


Bigger radius
Should not be
tessellated

- Suppose a model with overall dimensions of 250x250x250 mm has a round corner with a radius of 1 mm.
- the results of tessellating the model by applying Chord Heights of 0.5 and 0.05 mm respectively as shown in figure.
- Unfortunately, the **decrease of chord height** leads not only to **smoother surfaces** but also to **larger data files**.
- Therefore, a compromise parameter value should be selected to obtain the best trade-off between **accuracy** and **file size**.



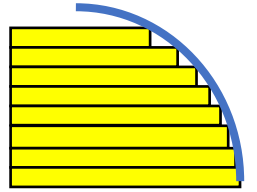
Chord Heights of 0.5
Small size



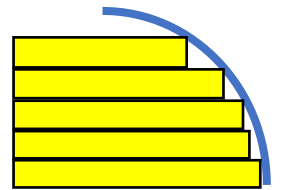
Chord Heights of 0.05 mm
Large file

ERRORS DUE TO SLICING

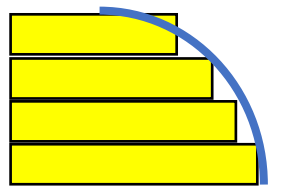
- RP processes have a stair-stepping problem that is found in all layer manufacturing technologies.
- **Stair – stepping** is a consequence of the **addition of material in layers**.
- As a result of this discrete layering, the **shape** of the original CAD models in the build direction (z) is **approximated** with **stair-steps**.
- This type of **error** is due to the **working principles of RP** processes, which can be assessed in data preparation.



fine layer
thickness



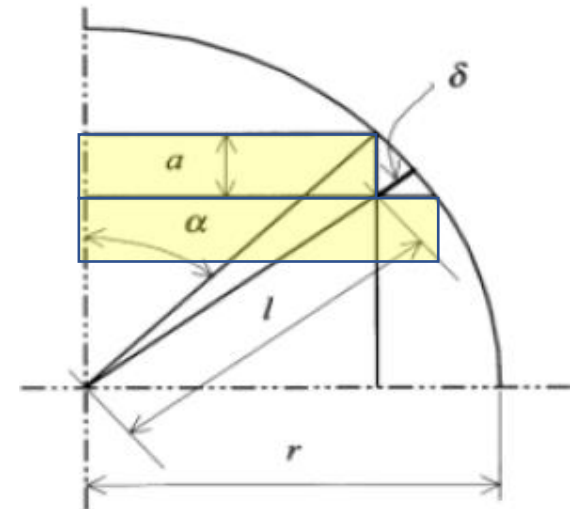
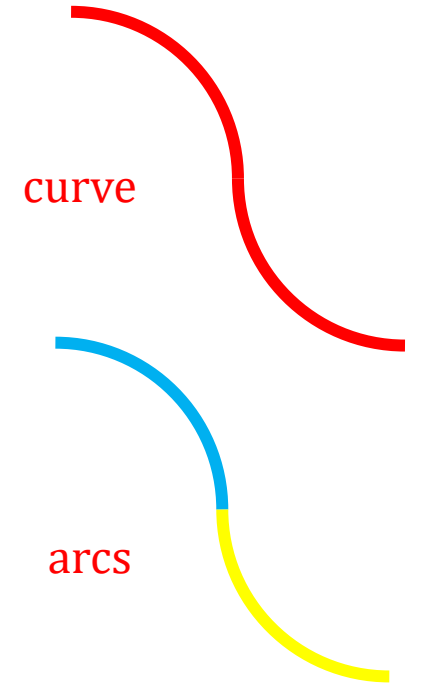
medium layer
thickness



Large layer
thickness

- Mathematically, curves are described with curvatures and a curvature radius.
- **curves** in a section of a CAD model can be replaced with **arcs**. To assess the error of stair-steps, arcs can also be used.
- The **error** due to the replacement of a circular arc with stair-steps is illustrated in Figure and defined as:

$$\delta = r - l$$



Part Building Errors

The main Type of Errors in Building of Parts by SLS and SGC process are:

1. Curing Errors

Errors Caused by over curing and scanned line shape.

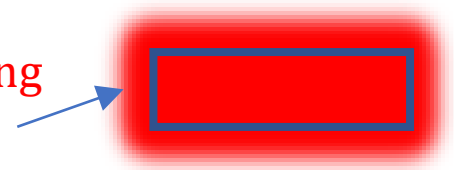
2. Control Errors

Errors caused by layer thickness and scan position control.

- **Over curing:** Laser over curing is necessary to adhere layers to form solid parts.

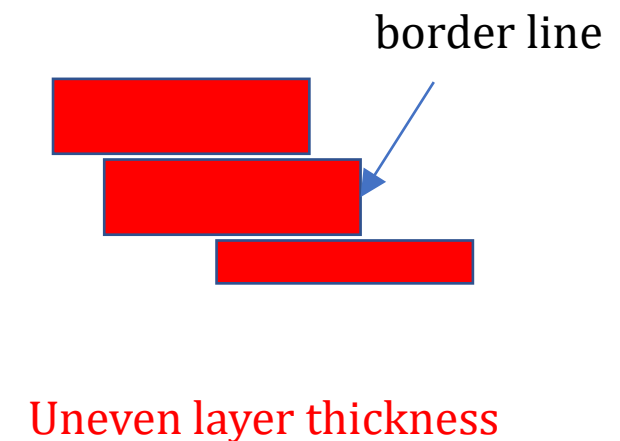
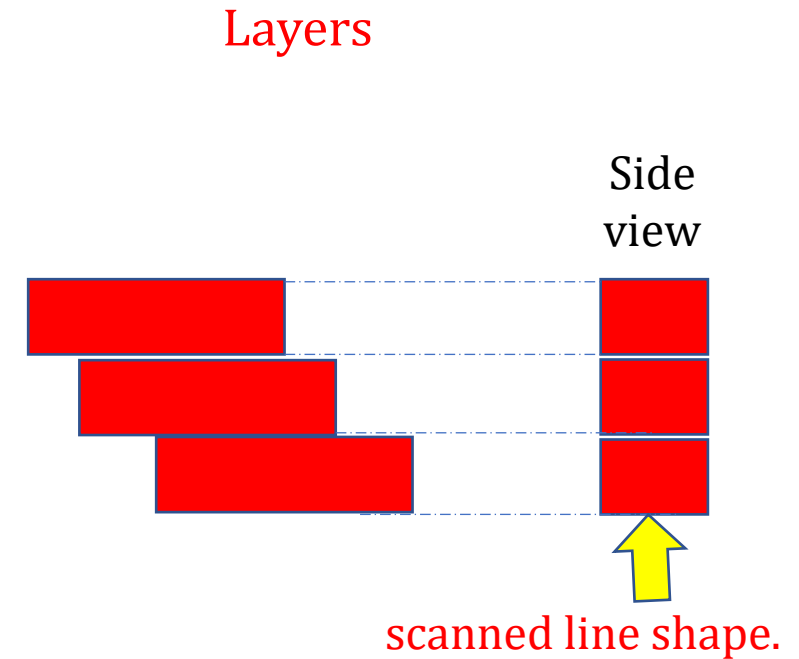
Over curing causes dimensional and positional errors.

Error: Basically curing
outside of what is
needed



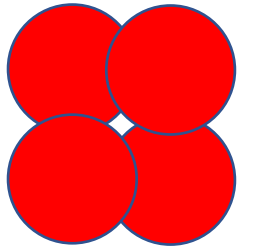
Scanned line shape: A scanned line is created when a laser beam scans the resin surface. The cross section of the scanned line is referred to as the scanned line shape.

- **Control errors:** Theoretically, the layer thickness should be at the defined value and the border line should be positioned at the specified positions. The layer thickness is variable and the border position is not precise.

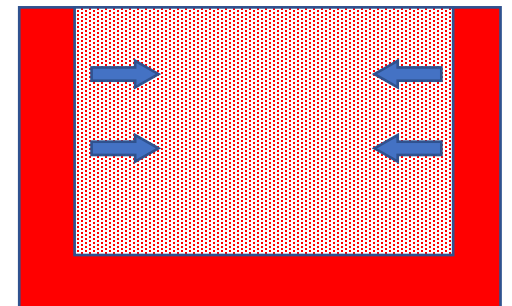
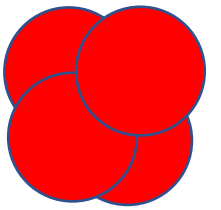


Part building errors in the SLS process

- The main cause of part inaccuracy is the **shrinkage** during sintering which **does not** always occur in a **uniform manner**.
- The shrinkage of a new layer can be constrained by the existing part substrate or by support powder trapped within enclosed areas.
- In addition, areas at **high temperatures** tend to **shrink more** than those at lower temperatures and part geometries such as **thick walls** or sections can **increase the shrinkage**.



Shrinkage
due to
cooling

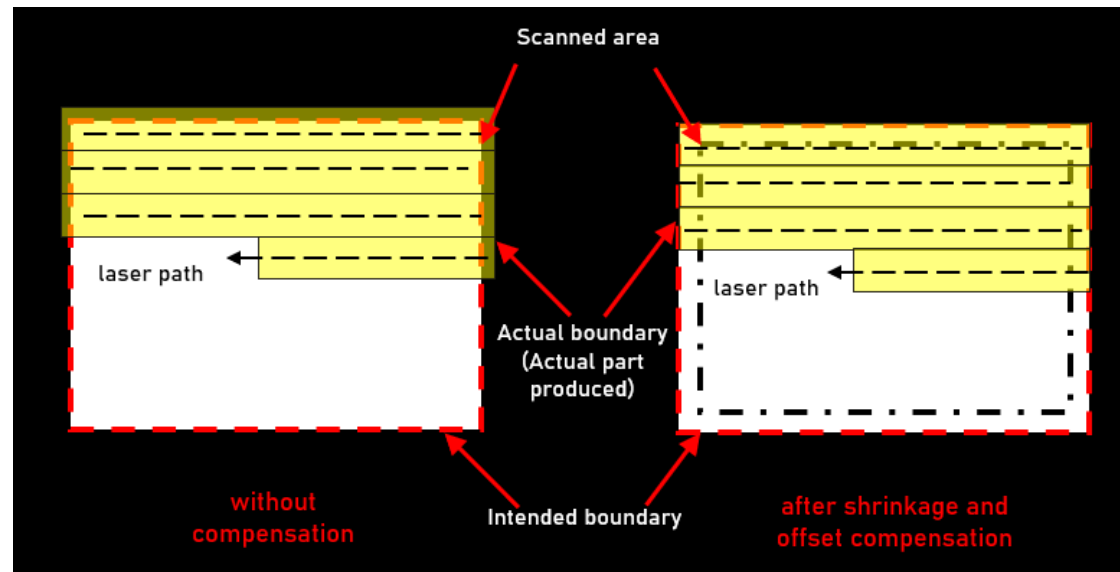


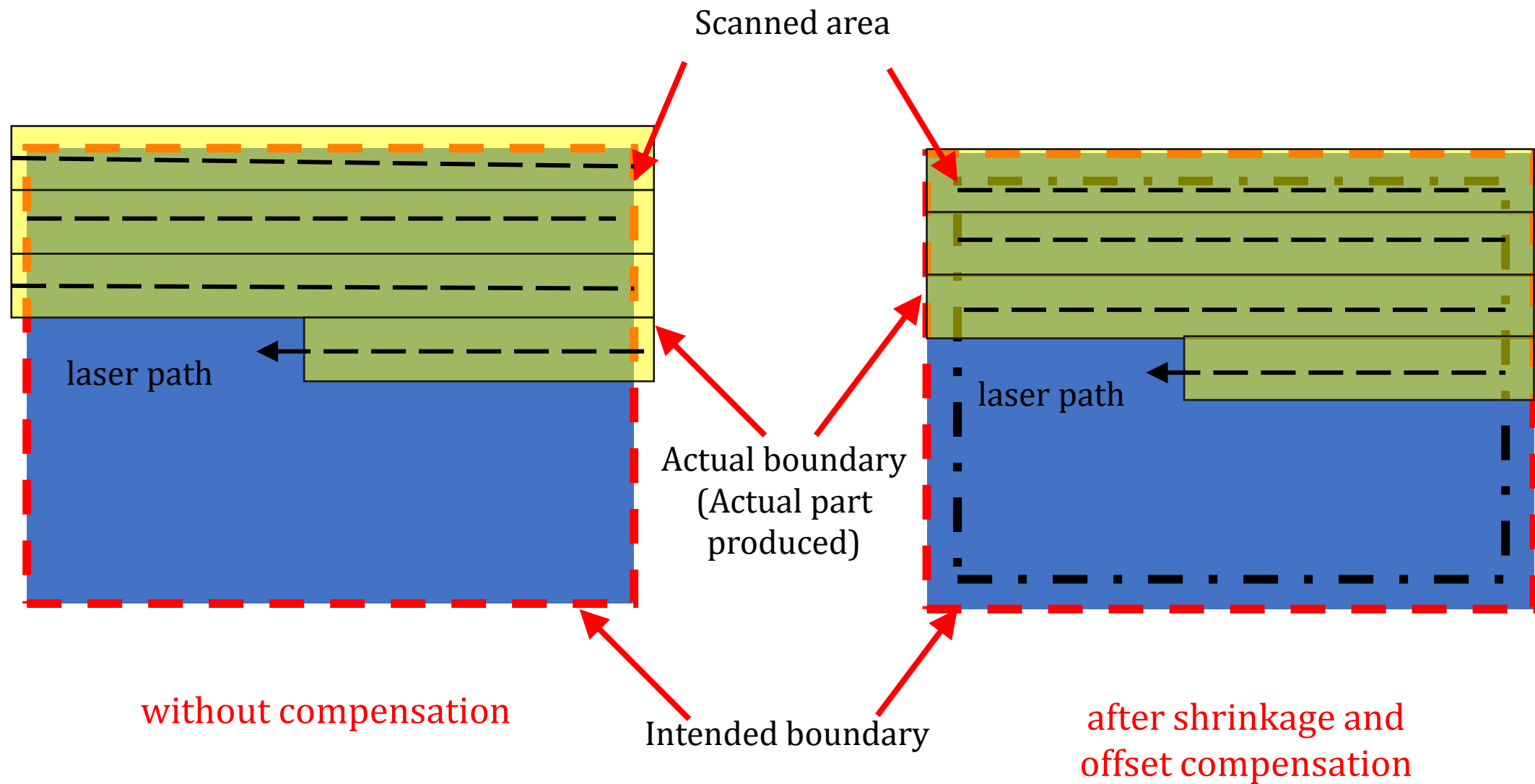
The shrinkage
constrained by
the support
powder

- To **compensate for shrinkage**, a material coefficient is calculated using a test part and a **scaling factor** is applied in each direction to the STL file.
- In practice, to **compensate** for the shrinkage, scaling and offsetting are applied to the part dimensions according to the following relation:

$$\text{New dimension} = a (\text{desired dimension}) + b$$

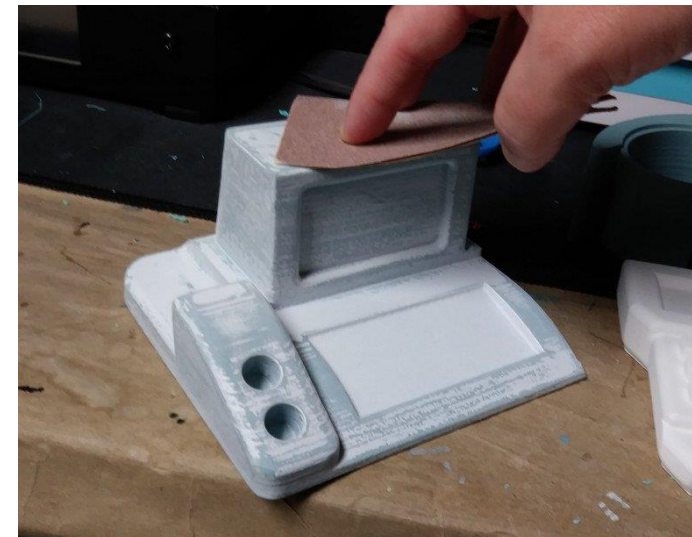
- In this way, it is possible to compensate for the shrinkage occurring during the SLS process and for the part growth due to the laser beam melting diameter (figure).





PART FINISHING

- Some RP applications such as fabrication of exhibition quality models, tooling or master patterns for indirect tool production require additional finishing improving the surface appearance of the part.
- To achieve this, the stair-step effect on important surfaces has to be removed. Usually, this is done by **sanding** and **polishing** RP models, which leads to changes in feature shapes, dimensions and positions.



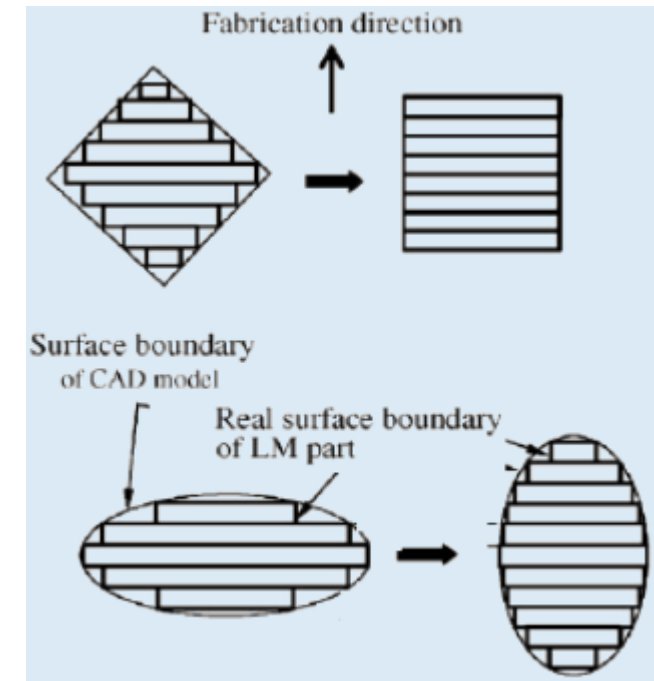
- The model accuracy after finishing operations is influenced mostly by **two factors**,
 1. **the varying amount of material that has to be removed** and
 2. **the finishing technique adopted.**
- These two factors determine to what extent the dimensional accuracy of RP models will be reduced during finishing.

Varying amount of Material

During the data preparation stage, the RP model shapes are approximated with the corners of stair-steps.

Each RP process reproduces the **corners** as the stair-steps with **different resolution**.

Hence, the **amount of material** that has to be removed to improve the surfaces of the same model can **vary** due to the selected **part build orientation**.



Different part build orientation.

Finishing Technique

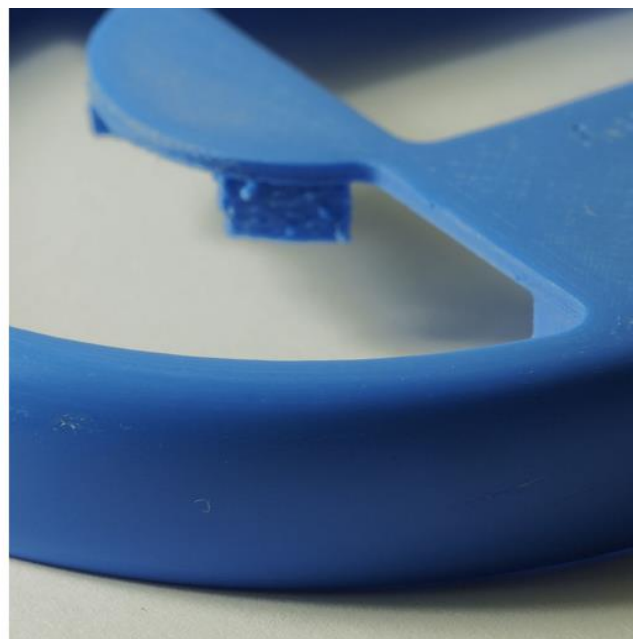
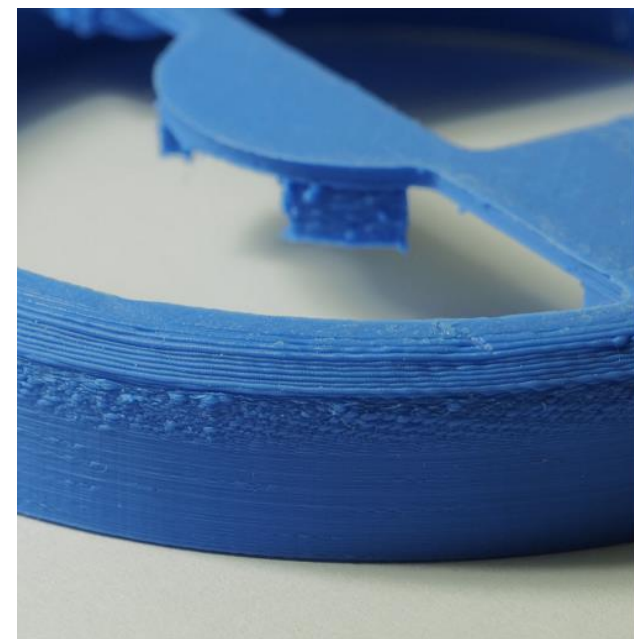
A number of processes can be employed to finish RP models

1. **wet and dry sanding**
2. **sand blasting**
3. **Coating**
4. **Spraying**
5. **Infiltration with special solutions**
6. **Machining *and many more.***

Techniques with better dimensional control will have less impact on model accuracy.

Models finished employing **milling** will have **less influence** on **accuracy** than those using **manual wet sanding** or **sand blasting**.





fresh print



one coat



two coats



three coats



INFLUENCE OF BUILD ORIENTATION

1. Selection of part build orientation
2. Orientation constraints of the sla process
3. Orientation constraints of the sls process

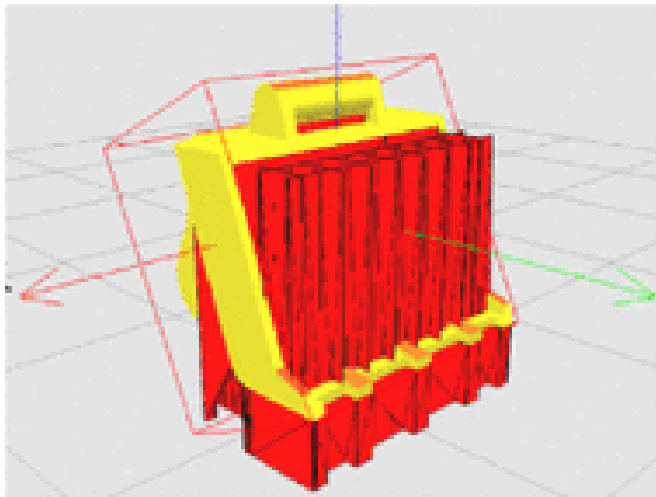
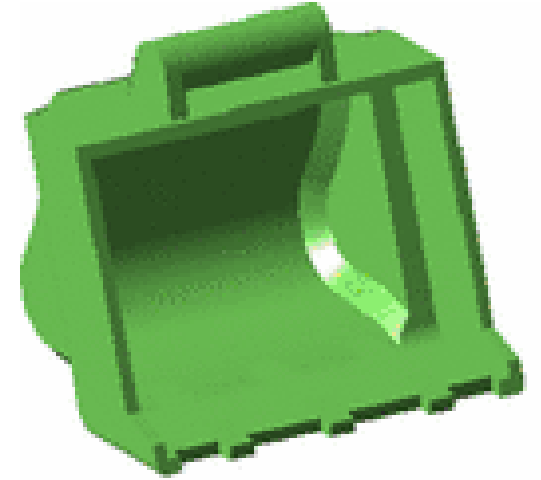
How will you
orient this part??



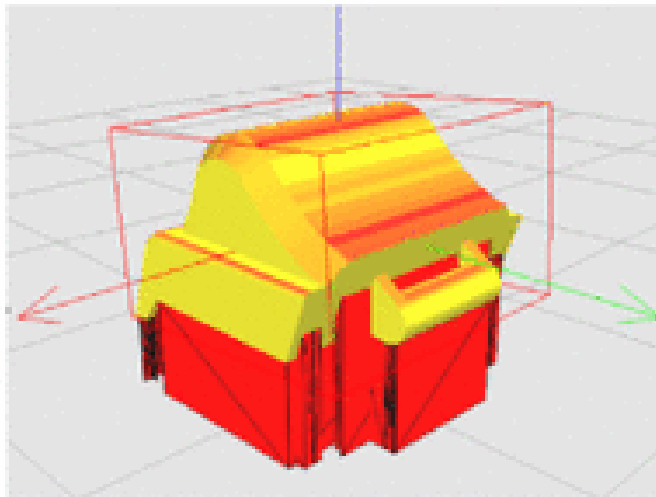
SELECTION OF PART BUILD ORIENTATION

The selection of the part build orientation is a very important factor.

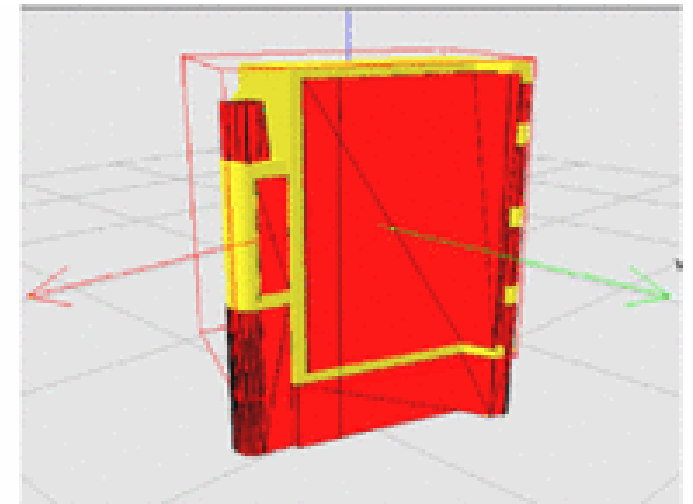
It results in **minimizing build time and costs** and achieving optimal **accuracy**.



Practical orientation 1



Practical orientation 2



Practical orientation 3

SELECTION OF PART BUILD ORIENTATION

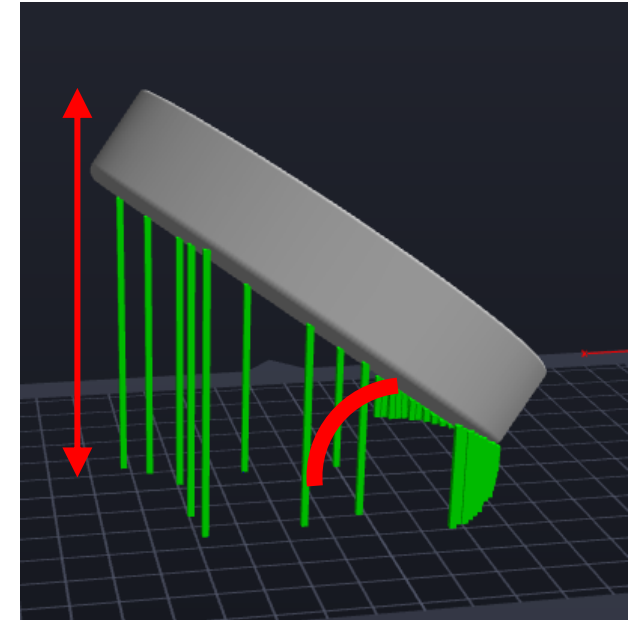
The build should be oriented to:

1. **Minimize** the distance between the **part** and **CAD origin**.
2. **Minimize** the **height of the support structure**.
3. **Minimize** the **overall height** of the object

This **reduces** the **number of layers** to be built, thereby reducing the build time.

4. **Minimize** the number of **slanted or sloping surfaces**.

Choosing the best orientation is a **multi-criteria task** that involves trade-offs between **maximizing the surface smoothness and accuracy** of important features and **minimizing the build time and cost**.



ORIENTATION CONSTRAINTS OF THE SL PROCESS

The constraints are defined by:

1. Technological capability
2. Z direction
3. Necessity for support structures for **overhanging** areas.

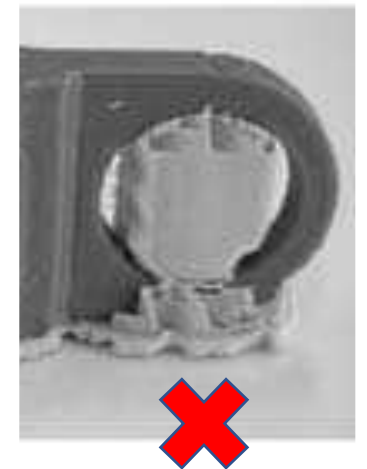
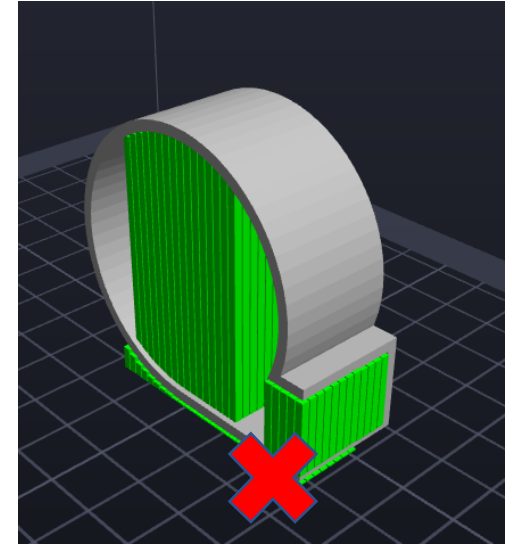


Following are features constraints SL process:

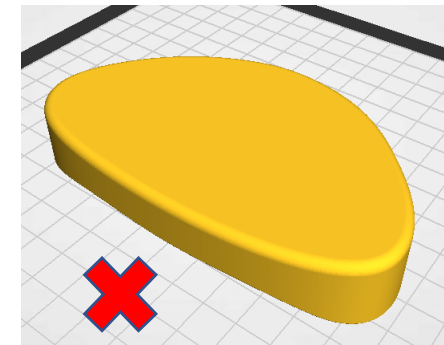
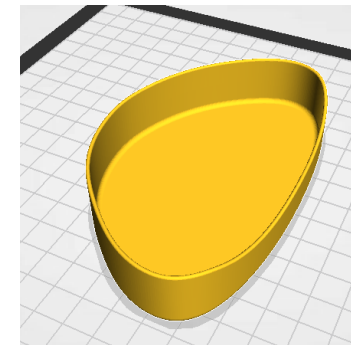
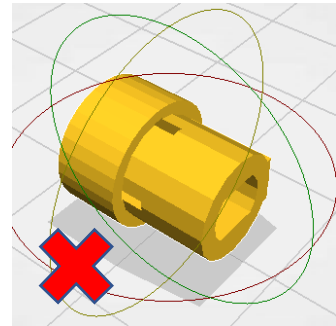
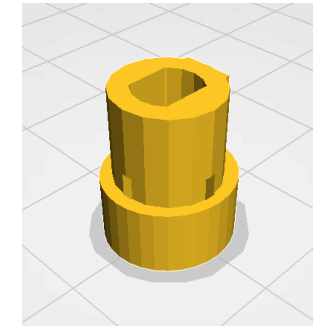
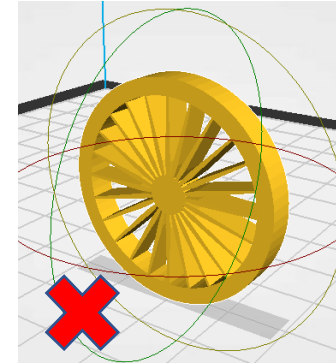
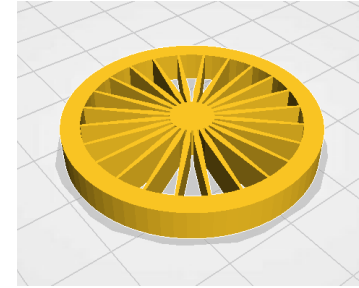
Users specified critical surfaces: Plane Surfaces, have to be placed with their normal along the build directions *i.e.*, they are horizontal and upward facing.

Coordinate systems: Created & employed by the designer during modeling. The orientation of the coordinate axes represent logical build orientation.

Holes: To avoid hard removing of supports and stair-stepping inside holes, these are placed orthogonally to horizontal planes.



- **Cuts:** If these curve through the part entirely or have a depth greater than a certain minimum, the planes which they cut through are made horizontal otherwise they are ignored.
- **Shafts:** These are oriented such that their placement planes are horizontal in order to provide the best **external surface finish**.
- **Protrusions:** These are created by revolving a section, the axes are positioned. So that they are **vertical**.
- **Shells:** These are orientated so that the **concave** part of the shell faces upwards in order to minimize internal supports.
- **Axes:** All the axes are placed so that they are vertical.



ORIENTATION CONSTRAINTS OF THE SLS PROCESS

The orientation constraints of the SLS process are dictated mostly by two factors:

1. The process accuracy in the x - y plane and z direction
2. Material used, since the shrinkage and anisotropic properties of the sintered powders are different.

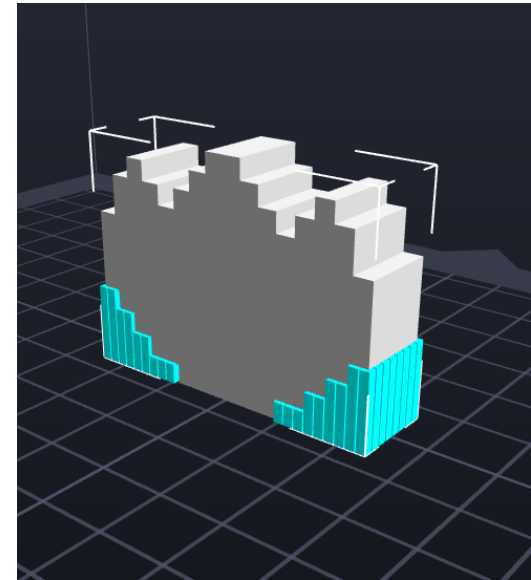
The SLS material falls into the following categories.

1. Polycarbonates
2. Polyamides
3. Elastomers
4. Metals
5. Sands

ORIENTATION CONSTRAINTS OF THE SLS PROCESS

Different materials impose different orientation constraints on design features.

- To **maximize strength**, parts built in the x-y plane so that they will have improved flexibility.
- To have best feature definition, critical features should be built on **upwards facing model surfaces**.
- **Rate of change of cross sectional is not too rapid** to ensure consistent shrinkage. Thus, parts are normally **angled by 10-15 degree** around x-axes. After an orientation has been chosen, the user is advised to slice the part to check that this rate of change is acceptable.
- **Build the prototype in the same orientation** as that in which it will be **cured to reduce warping**. Therefore, part must be stable in the build orientation.



ORIENTATION CONSTRAINTS OF THE SLS PROCESS

- Thin **walls** should be built **perpendicular** to the **rollers direction** of travel to avoid their distortion when building with polycarbonates and elastomers.
- Shells should be built so that the **concave** side points **upwards** to reduce trapped heat and minimize growth. This is done when building with polycarbonates.
- **Overhangs** should be **minimized** when building with polycarbonates, sand and elastomers.
- When making metal tooling inserts, the build direction should be chosen such that **the parting line of the mould is built facing upwards.**

